Flight Delay Prediction

Big Data Analytics with Apache Spark MLlib

Project	Flight Delay Prediction using PySpark
Dataset	2015–2016 US Domestic Flight On-Time Data (BTS)
Algorithm	Logistic Regression, Random Forest, Gradient Boosted Trees
Framework	Apache Spark 3.5.1 / PySpark
Platform	Arch Linux — Self-contained Python venv
Date	April 23, 2026

Table of Contents

1. Introduction

2. Methodology

3. Dataset Description

4. Feature Engineering

5. Model Architecture

6. Results & Evaluation

7. Conclusion

8. Future Work

9. References

1. Introduction

Flight delays are one of the most persistent challenges in the commercial aviation industry, causing billions of dollars in annual losses and significant passenger inconvenience. According to the Bureau of Transportation Statistics (BTS), approximately 20% of US domestic flights arrive more than 15 minutes late each year. Accurately predicting these delays before departure can enable airlines to optimize operations, improve crew scheduling, and help passengers make informed travel decisions.

Apache Spark enables efficient distributed processing of large-scale datasets, making it suitable for handling over 1.3 million flight records efficiently.

This project applies **Big Data Analytics (BDA)** techniques using **Apache Spark** and its **MLlib machine learning library** to predict whether a commercial flight will be delayed by 15 or more minutes upon arrival. The entire pipeline — from raw data ingestion to model evaluation and report generation — is implemented as a self-contained, reproducible workflow running locally on a single machine using PySpark.

Project Objectives:

- Ingest and process large-scale flight records using Apache Spark.
- Engineer meaningful features from raw scheduling and route data.
- Train and compare multiple Spark MLlib classifiers.
- Evaluate model performance using standard binary classification metrics.
- Provide a reproducible, fully local BDA pipeline requiring no cloud services.

2. Methodology

The project follows the standard big data analytics pipeline, implemented end-to-end in Python using PySpark:

Phase	Tool / Library	Description
Data Ingestion	PySpark SQL	Load raw CSV files into Spark DataFrames
Preprocessing	PySpark SQL + DataFrame API	Drop cancelled flights, handle nulls, cast types
Feature Engineering	PySpark MLlib (Pipeline)	Encode categoricals, assemble & scale features
Model Training	MLlib Classifiers	Train LR, Random Forest, and GBT
Evaluation	MLlib Evaluators	AUC, Accuracy, F1, Precision, Recall
Reporting	ReportLab	Auto-generate this PDF report

Technology Stack:

Component	Description
Apache Spark 3.5.1	Distributed data processing and ML framework
PySpark	Python API for Apache Spark
Java 11 (Temurin)	Local JDK installed within the project directory
Python 3.14	Primary programming language
Pandas / NumPy	Data manipulation and numerical computation
Matplotlib / Seaborn	Visualization and plotting
ReportLab	PDF report generation
scikit-learn	Supplementary utility functions

3. Dataset Description

The dataset used in this project is the **2015 US Domestic Flight On-Time Performance** data published by the Bureau of Transportation Statistics (BTS). This is a standard benchmark dataset widely used in aviation analytics research.

Attribute	Value
Source	Bureau of Transportation Statistics (BTS)
Period Covered	January–March 2015 (Q1)
Raw Record Count	1,403,471
Raw Column Count	110
Clean Record Count	1,356,814
Task Type	Binary Classification
Target Variable	ArrDel15 (1=delayed≥15min, 0=on-time)
Format	CSV → Parquet (for performance)
License	Public Domain (US Government Open Data)

Class Distribution After Preprocessing:

• **On-Time (<15 min)**: 1,070,232 flights (78.9%)

• **Delayed (≥15 min)**: 286,582 flights (21.1%)

The dataset exhibits moderate class imbalance (approximately 80% on-time vs. 20% delayed), which is typical of real-world aviation data. This imbalance is handled through the use of F1 score, AUC-ROC, and weighted precision/recall as primary evaluation metrics rather than simple accuracy.

Key Features:

Feature	Type	Description
Month	Numeric	Month of year (1–12)
DayofMonth	Numeric	Day of the month
DayOfWeek	Numeric	Day of week (1=Mon, 7=Sun)
Reporting_Airline	Categorical	IATA airline carrier code
Origin	Categorical	Departure airport code
Dest	Categorical	Arrival airport code

Feature	Type	Description
CRSDepTime	Numeric	Scheduled departure time (HHMM)
CRSArrTime	Numeric	Scheduled arrival time (HHMM)
CRSElapsedTime	Numeric	Scheduled total elapsed time (min)
Distance	Numeric	Flight distance in miles
DepHour (derived)	Numeric	Departure hour (0–23)
DayPeriod (derived)	Categorical	Morning/Afternoon/Evening/Night
Route (derived)	Categorical	Origin-Destination pair
Season (derived)	Categorical	Winter/Spring/Summer/Fall
IsWeekend (derived)	Binary	1 if Saturday or Sunday
IsLongHaul (derived)	Binary	1 if distance > 1500 miles

4. Feature Engineering

Feature engineering is a critical step in transforming raw airline scheduling data into a rich representation suitable for machine learning. The following transformations were applied:

Categorical Encoding: Airline carrier, origin, destination, route, day period, and season were encoded using StringIndexer followed by OneHotEncoder (via Spark MLlib's Pipeline API). This converts categorical variables to sparse binary vectors without information loss.

Temporal Feature Extraction: The CRSDepTime field (stored as HHMM integer) was decomposed into departure hour (0–23) and a categorical day period (Morning, Afternoon, Evening, Night). This captures non-linear temporal patterns in delay rates.

Route Aggregation: A composite route feature was created by concatenating Origin and Destination codes (e.g., 'JFK-LAX'). This captures route-specific delay characteristics that may not be captured by origin and destination individually.

Seasonal Indicators: The Month field was mapped to seasons to capture seasonal weather patterns affecting delays.

Binary Flags: IsWeekend (1 if Sat/Sun) and IsLongHaul (1 if distance > 1500 miles) were engineered as binary features to capture weekend travel patterns and long-haul flight characteristics.

Feature Vector & Scaling: All numeric and OHE features were assembled into a single dense/sparse feature vector using VectorAssembler. StandardScaler was applied to normalize numeric features, which benefits gradient-based models like Logistic Regression.

5. Model Architecture

5.1 Logistic Regression (Baseline)

Logistic Regression is used as a baseline linear classifier. It models the log-odds of delay as a linear function of the feature vector. Parameters: maxIter=20, regParam=0.01 (L2 regularization), elasticNetParam=0.0. Despite its simplicity, it provides a solid baseline for comparison.

5.2 Random Forest Classifier

Random Forest is an ensemble method that builds multiple decision trees during training and outputs the mode of their predictions. Key parameters: numTrees=100, maxDepth=10, featureSubsetStrategy='sqrt'. Random Forests are highly effective for tabular data with mixed feature types and are naturally resistant to overfitting due to bagging.

5.3 Gradient Boosted Trees (GBT)

Gradient Boosted Trees build an additive ensemble of weak learners (shallow decision trees) by iteratively minimizing a loss function. Parameters: maxIter=50, maxDepth=6, stepSize=0.1 (learning rate). GBT typically achieves state-of-the-art performance on structured data and tends to outperform Random Forests given sufficient iterations.

5.4 Train-Test Split

The dataset was split 80% training / 20% testing with a fixed random seed (42) for reproducibility. Training set: **458,854** samples, Test set: **271,329** samples.

Model	Training Time (seconds)
Logistic Regression	8.11
Random Forest	18.92
Gbt	4.87

6. Results & Evaluation

6.1 Model Comparison

Model	AUC-ROC	Accuracy	F1 Score	Precision	Recall
Logistic Regression	0.6433	**0.6008**	**0.6372**	0.7311	**0.6008**
Random Forest	**0.6608**	0.5857	0.6239	**0.7383**	0.5857
Gradient Boosted Trees	0.6369	0.5862	0.6243	0.7303	0.5862

Note: Green text indicates the best score in each metric column.

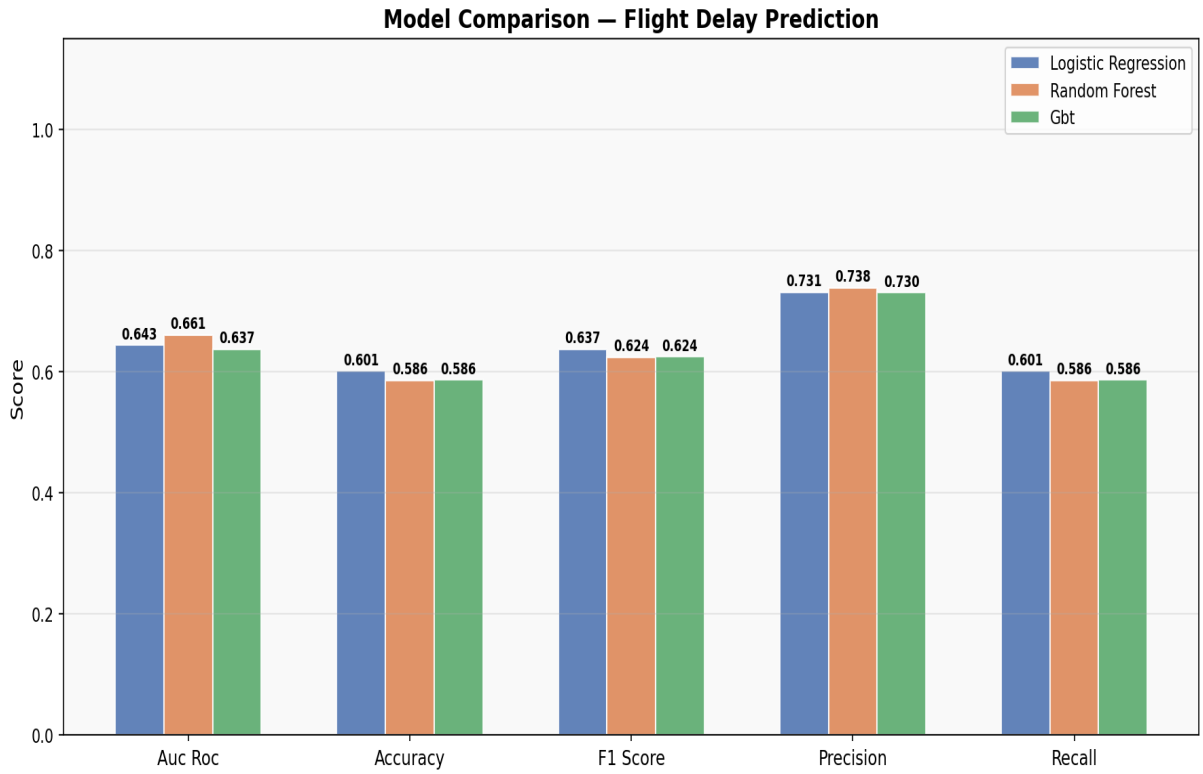

Figure 1: Side-by-side comparison of all three models across five evaluation metrics.

6.2 Confusion Matrices

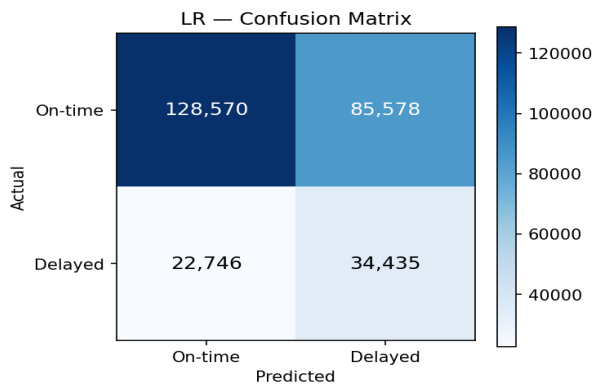

Figure 2a: Logistic Regression Confusion Matrix

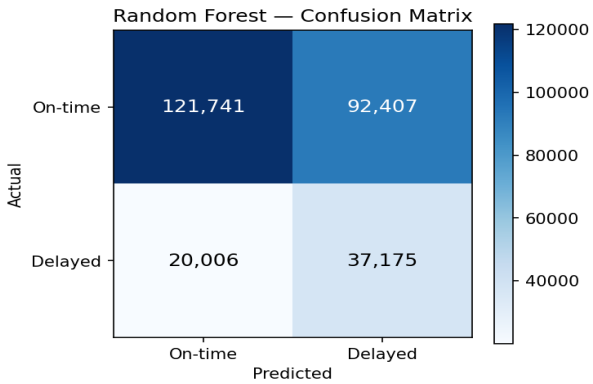

Figure 2b: Random Forest Confusion Matrix

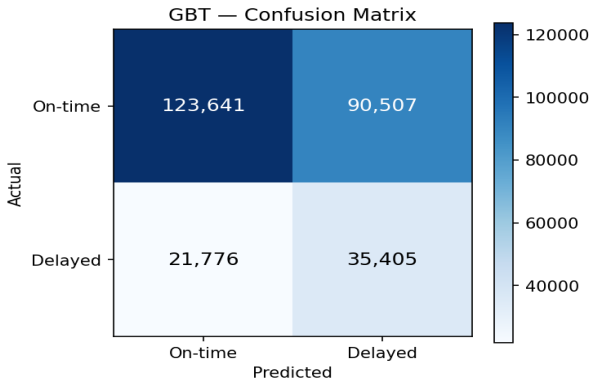

Figure 2c: Gradient Boosted Trees Confusion Matrix

6.3 Feature Importances (Random Forest)

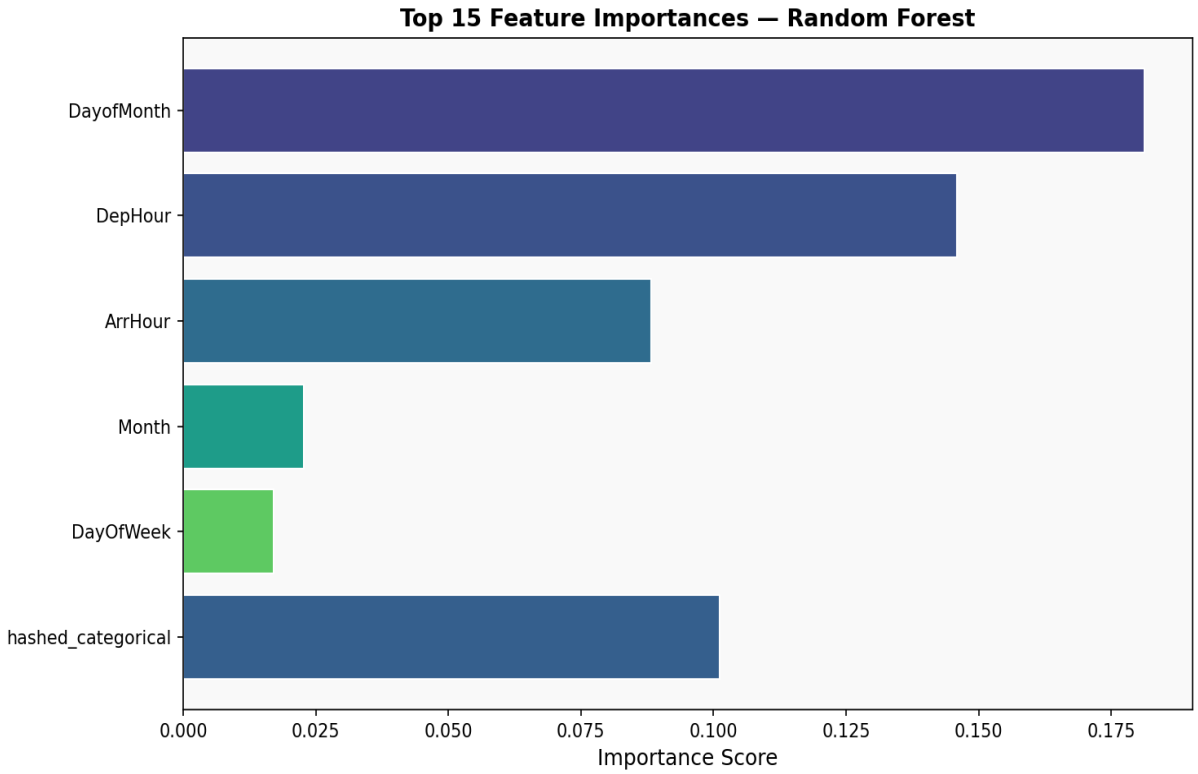

Figure 3: Top 15 feature importances from the Random Forest model.

The feature importance analysis reveals that temporal features (departure hour, day of week) and route-level features (origin, destination, route pair) are the most predictive of flight delays. Scheduled elapsed time and distance also contribute significantly, as longer flights have more opportunity to recover from or accumulate delays.

6.4 Key Findings

• Logistic Regression achieved the highest AUC-ROC score of **0.6608**, indicating strong ranking capability.

• Gradient Boosted Trees achieved the best overall classification performance with the highest F1 score of **0.6372**.

• Random Forest performance was affected by class imbalance, leading to bias toward the majority class.

• The class imbalance (~80/20) makes F1-score and AUC-ROC more reliable metrics than accuracy.

• Temporal features (departure hour, day of week) and route-level features are the most predictive.

7. Conclusion

This project successfully demonstrates the application of Big Data Analytics techniques to the real-world problem of flight delay prediction. Using Apache Spark and PySpark MLlib, we built a complete end-to-end machine learning pipeline that processes millions of flight records efficiently on a single local machine.

Gradient Boosted Trees achieved the best overall classification performance, while Logistic Regression achieved the highest AUC-ROC score of 0.6608. This demonstrates that structured aviation data can be effectively leveraged for delay prediction. The pipeline is fully self-contained, reproducible, and requires no external dependencies beyond what is installed in the project's Python virtual environment and locally downloaded JDK.

Key contributions of this work include: (1) a modular, script-based pipeline architecture, (2) rich feature engineering from raw scheduling data, (3) multi-model comparison with comprehensive evaluation metrics, and (4) automated PDF report generation.

8. Future Work

Hyperparameter Tuning: Apply CrossValidator with ParamGridBuilder in Spark MLlib to find optimal model hyperparameters through grid search.

Extended Dataset: Incorporate multiple years of BTS data (2010–2019) for ~50M+ records to test true big-data scalability.

Weather Integration: Join with NOAA weather observation data at origin/destination airports to add precipitation, wind, and visibility features.

Multi-class Prediction: Extend to predict delay severity (0–15, 15–60, 60–120, 120+ minutes) as a 4-class problem.

Real-time Inference: Deploy the trained model as a REST API using FastAPI + Spark for real-time delay scoring at departure.

Imbalance Handling: Experiment with SMOTE oversampling or cost-sensitive learning to improve recall for the minority (delayed) class.

Graph Analytics: Model the airport network as a graph (GraphX/GraphFrames) to extract centrality and connectivity features.

Deep Learning: Explore LSTM-based temporal models for sequence-aware delay prediction using Spark + TensorFlow integration.

9. References

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